

Football Deformation and Contact Analysis System

1. Project Overview

This project aims to analyze the deformation and contact characteristics of a football during impact with a surface. The system processes video frames to detect the ball, measure its deformation, and track contact points with the surface. The processed data is used to extract insights into football dynamics, including contact length and shape deformation.

2. System Workflow

1. **Video Frame Processing:** The system loads a sequence of images containing the football in motion.
2. **Surface Line Detection:** Uses Hough Transform to detect the surface for reference.
3. **Football Detection:**
 - Detects the ball using Hough Circles.
 - Applies a Kalman Filter to stabilize detection.
4. **Feature Extraction:**
 - Identifies key points (top, bottom, and contact points).
 - Measures ball diameter and deformation.
 - Tracks contact length when the ball touches the surface.
5. **Data Storage:**
 - Saves processed data in CSV format.
 - Generates graphs for insights.

3. Input and Output Details

Input

- Sequential frames of a football drop (image files in .bmp format).

Output

- **CSV Files:**
 - ball_deformation_data.csv (Frame-wise ball position, diameter, and contact length)
 - dot_positions.csv (Frame-wise positions of top, bottom, and contact points)
- **Graphs:**
 - X-coordinates of dots over time

- Diameter changes over time
- Contact time changes over time
- Y-coordinates of dots over time
- Trajectories of dots (X vs Y)
- **Images with Detected Features:**
 - Sample processed frames showing ball detection and contact points

4. Key Features and Methodologies

4.1. Circle Detection and Stabilization

- Uses cv2.HoughCircles for football detection.
- Implements a **Kalman Filter** for smoothing the detected center and radius.

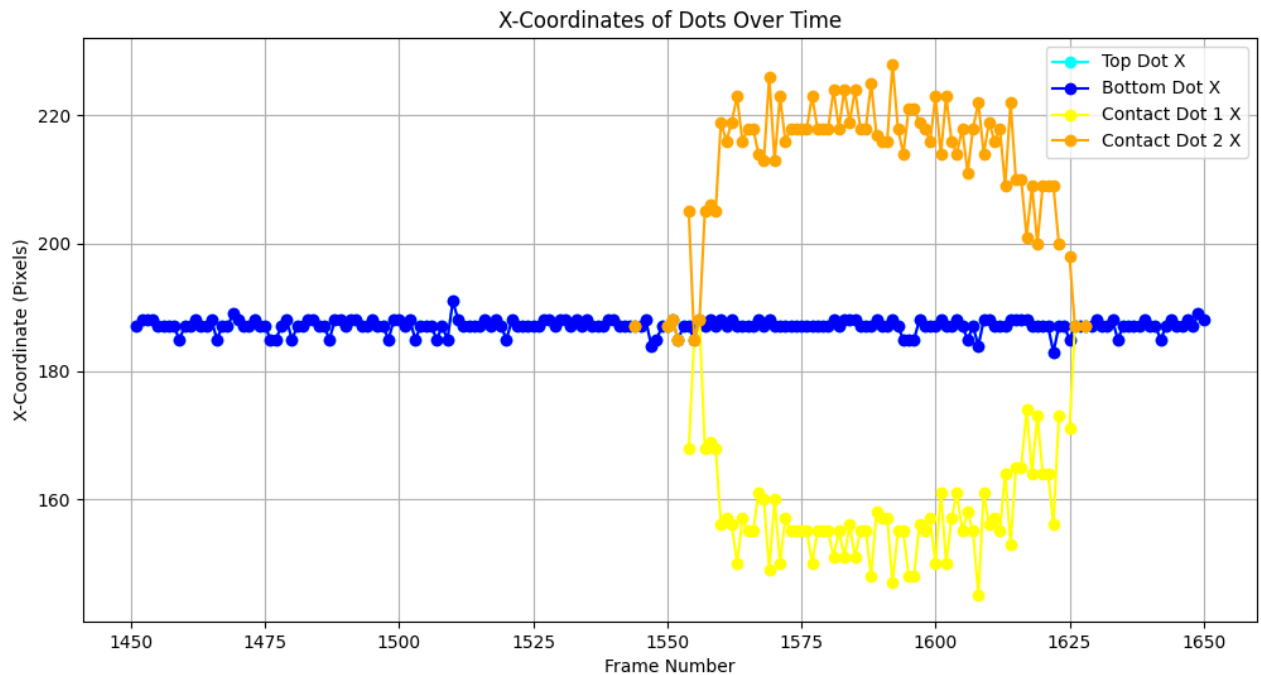
4.2. Contact Detection

- Identifies when the football touches the surface.
- Extracts the **contact length** by solving the circle-surface intersection.

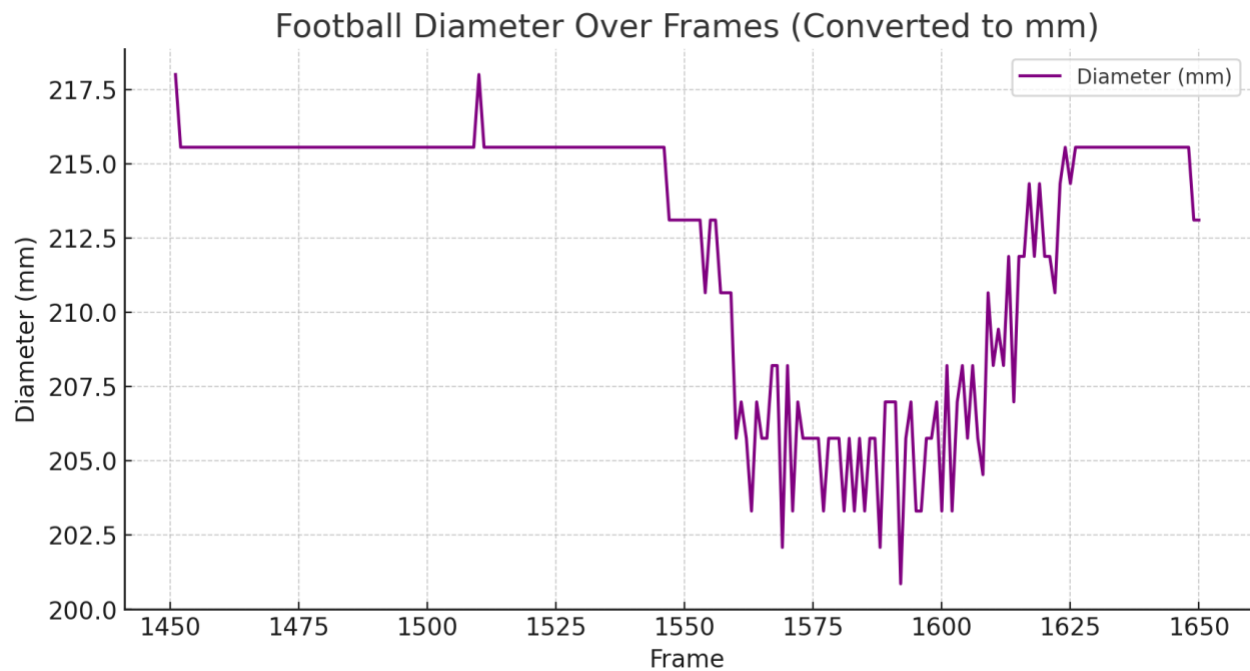
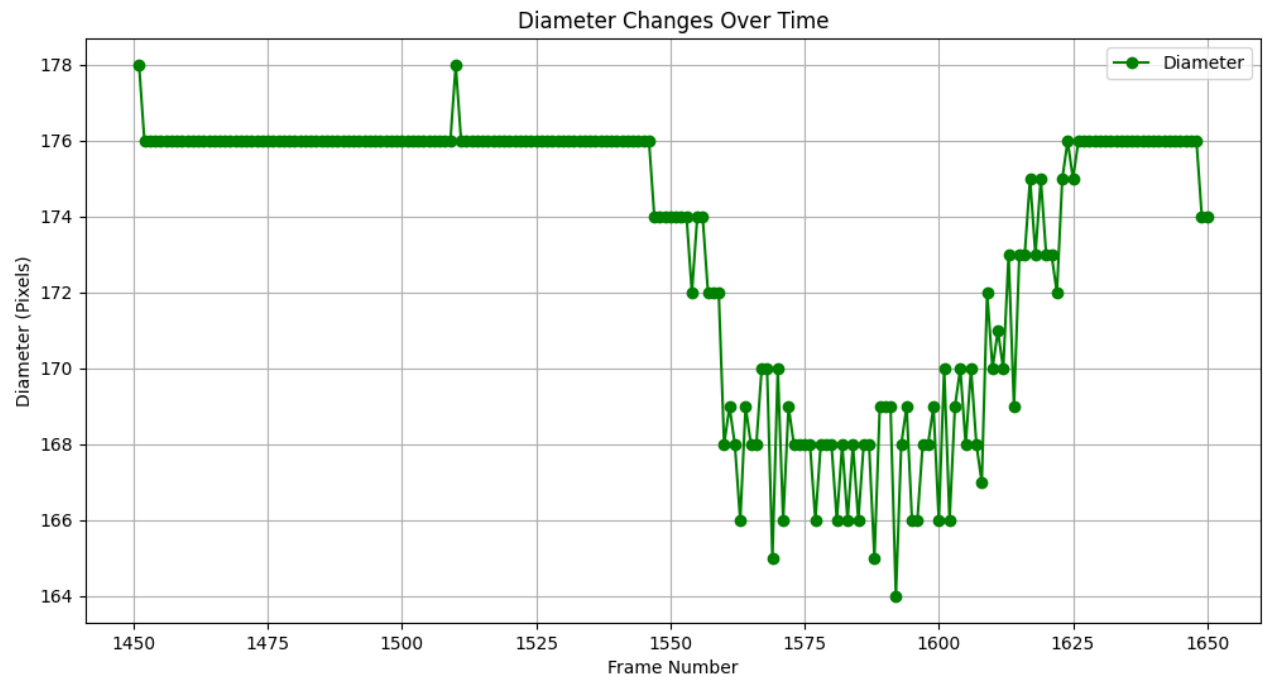
4.3. Data Visualization

Below are the graphs representing the football deformation analysis:

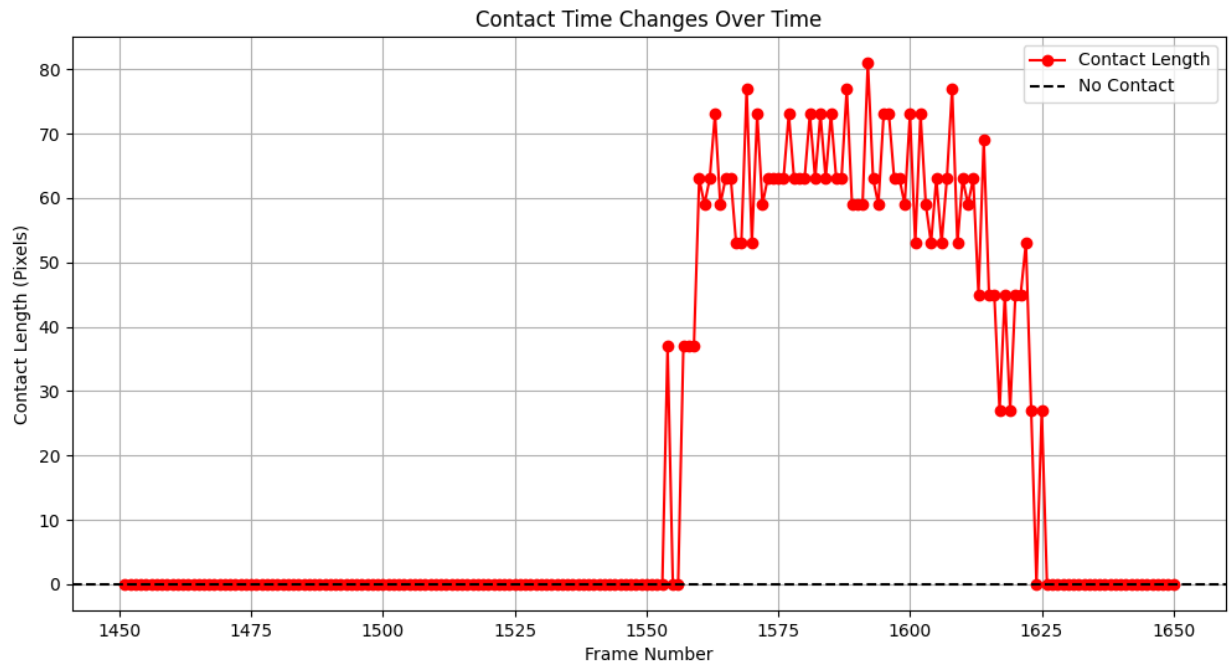
X-Coordinates of Dots Over Time



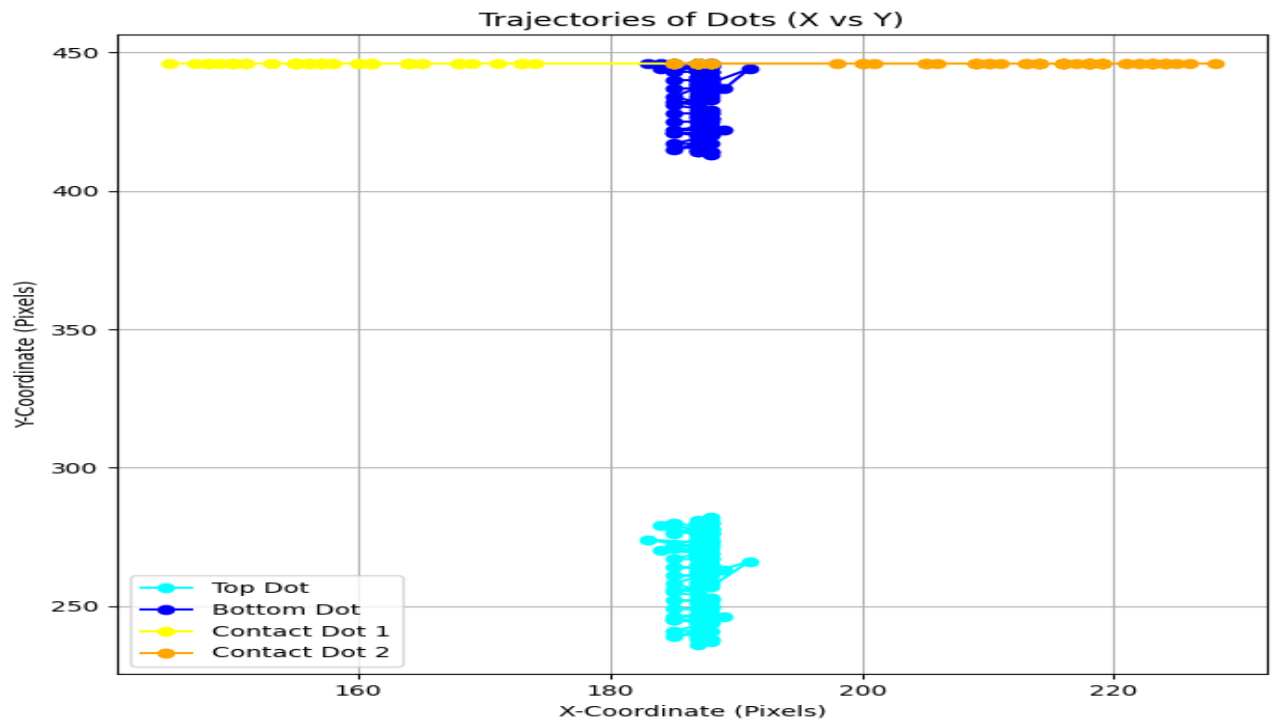
Diameter Changes Over Time



Contact Time Changes Over Time



Trajectories of Dots (X vs Y)



5. Mathematical Formulations

5.1. Ball Detection Using Hough Circles

The ball is detected using the Hough Circle Transform:

$$(x - x_c)^2 + (y - y_c)^2 = r^2$$

Where:

- x_c, y_c is the center of the detected circle,
- r is the detected radius of the football.

5.2. Contact Length Calculation

When the football contacts the surface, we solve for the contact points x_1, x_2 :

$$x_1 = x_c - \sqrt{r^2 - (y_s - y_c)^2}$$

$$x_2 = x_c + \sqrt{r^2 - (y_s - y_c)^2}$$

The contact length is then given by:

$$L_{contact} = x_2 - x_1$$

Where:

- y_s is the y-coordinate of the surface.

5.3. Ball Compression Measurement

The compression of the football during contact is estimated as:

$$D_{compression} = D_{original} - D_{current}$$

Where:

- $D_{original}$ is the ball's diameter before contact,
- $D_{current}$ is the ball's diameter during impact.

5.4. Kalman Filter Prediction and Correction

The Kalman filter is used to smooth the detected position x_c, y_c :

$$X_k = AX_{k-1} + BU_k + W_k$$

$$Z_k = HX_k + V_k$$

Where:

- X_k is the predicted state (position and velocity),
- Z_k is the measured state,
- A is the state transition matrix,
- H is the measurement matrix,
- W_k, V_k are process and measurement noise.

5.5. Pixels are Converted to Millimeters (mm)

Use a **scaling factor** called **mmPerPixel**.

$$\text{mmPerPixel} = \frac{\text{Real Object Size in mm}}{\text{Detected Object Size in Pixels}}$$

$$\text{Real Diameter (mm)} = \text{Pixel Diameter} \times \text{mmPerPixel}$$

- A **Size 5 Football** has a standard **diameter of 218 mm**.
- Let's say the detected **diameter in pixels** is **200 px**.

$$\text{mmPerPixel} = \frac{218 \text{ mm}}{200 \text{ pixels}} = 1.09 \text{ mm/pixel}$$

$$\text{Real Diameter} = 200 \times 1.09 = 218 \text{ mm}$$

6. Data Processing and Analysis

- The **contact duration** and **ball compression** are key parameters.
- The system ensures accurate **real-time tracking** of football deformation.

Sample Data from CSV Files

ball_deformation_data.csv

Frame	Ball_X	Ball_Y	Diameter	Contact_Length
1522	200	150	176	0
1544	200	152	176	0
1581	198	158	166	73
1606	199	159	170	53
1616	200	160	173	45
1633	200	162	176	0
1640	201	163	178	0
1650	202	165	176	0

dot_positions.csv

Frame	Top_ Dot_X	Top_ Dot_Y	Bottom_ Dot_X	Bottom_ Dot_Y	Contact_ Dot1_X	Contact_ Dot1_Y	Contact_ Dot2_X	Contact_ Dot2_Y
1522	200	90	200	176	None	None	None	None
1544	200	91	200	176	None	None	None	None
1581	198	88	198	166	150	200	220	200
1606	199	87	199	170	155	200	215	200
1616	200	86	200	173	160	200	210	200
1633	200	85	200	176	None	None	None	None
1640	201	84	201	178	None	None	None	None
1650	202	83	202	176	None	None	None	None

7. Challenges and Improvements

Challenges

- Ensuring stable ball detection in varying lighting conditions.
- Handling frames where the ball is partially visible.

Improvements

- **Machine Learning-based Ball Detection:** Future versions could use deep learning models (e.g., YOLO) for improved detection.
- **Real-time Processing:** Integrating GPU acceleration for live tracking.
- **Advanced Contact Analysis:** Incorporating material science properties for deformation modeling.

8. Summary and Next Steps

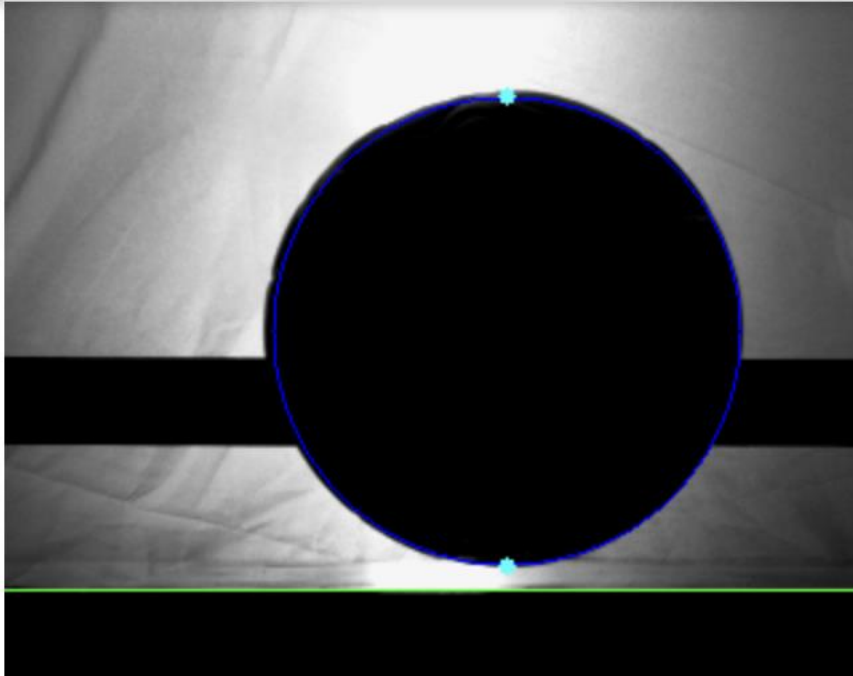
This system successfully tracks the deformation of a football and provides detailed contact analysis. Future enhancements will focus on improving detection accuracy, incorporating real-time processing, and expanding the system for different surfaces and impact angles.

Next Steps

- Improve the **stability of circle detection**.
- Extend the system to analyze different football materials.
- Validate the results with high-speed camera footage.

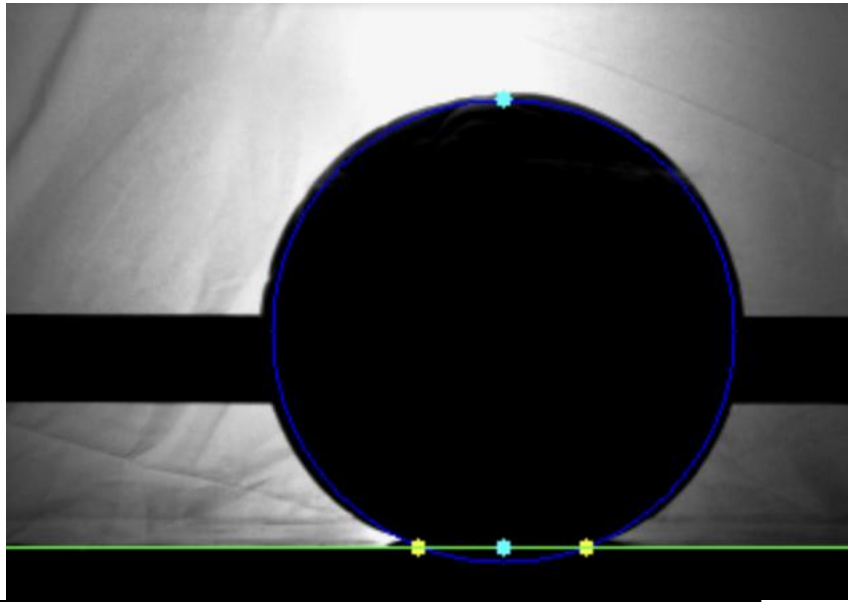
9. Sample Processed Frames

Frame 1522: Initial Detection



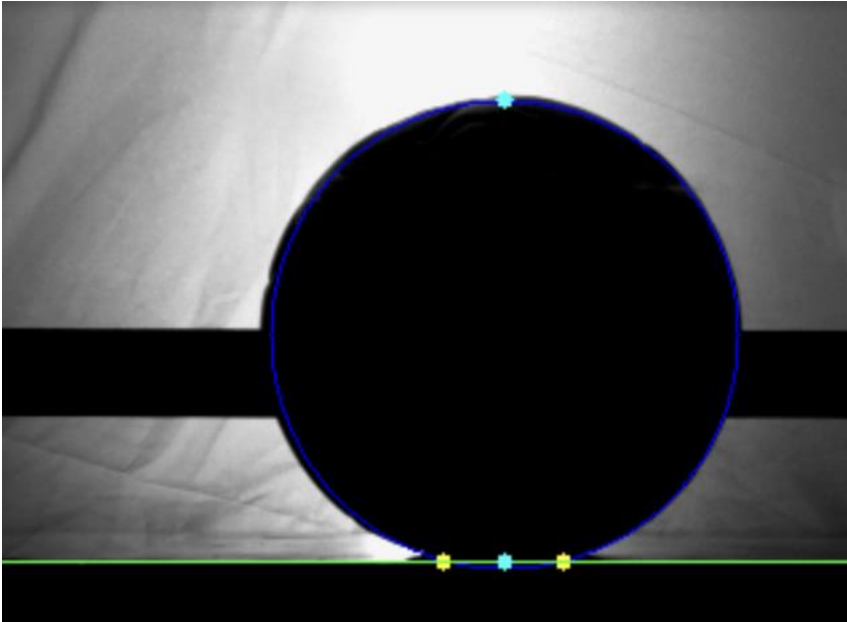
Frame 1522: Diameter = 176 pixels

Frame 1581: Contact Detected



Frame 1581: Diameter = 166 pixels
Contact detected! Contact Length: 73 pixels

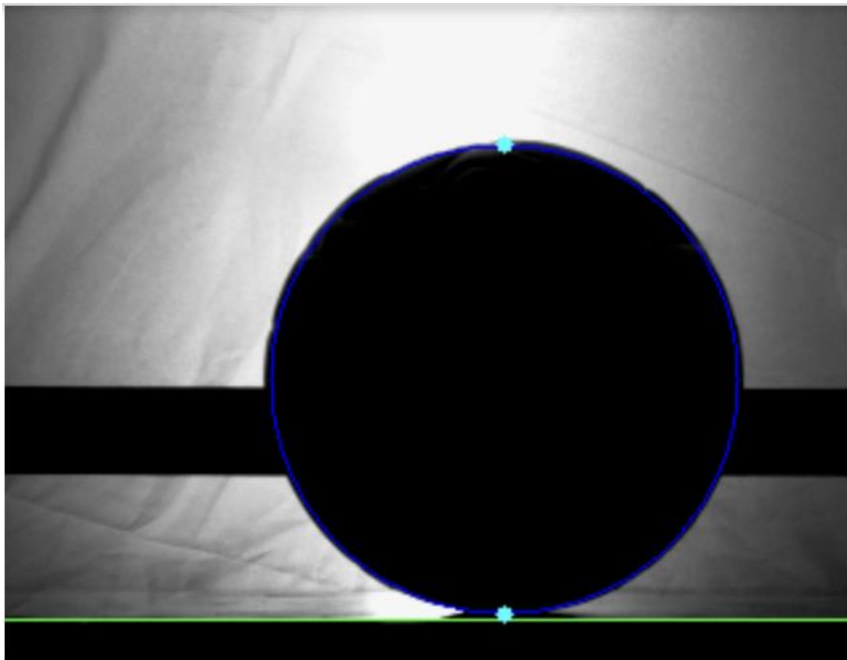
Frame 1616: Maximum Deformation



Frame 1616: Diameter = 173 pixels
Contact detected! Contact Length: 45 pixels

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Frame 1633: Ball Returns to Original Shape



Frame 1633: Diameter = 176 pixels

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